

DOT SENTINEL

AI POWERED CYBERSECURITY DASHBOARD

CAPSTONE REVIEW -PHASE II

GROUP 8



DOT SENTINEL

An AI-Powered Network Intrusion Detection & SOC Automation Platform

CAPSTONE PROJECT PHASE-2 | GROUP 08

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SENTINEL 

AI Powered Cyber-Security Dashboard



VIT BHOPAL UNIVERSITY | MARCH 2026

What is **DOT SENTINEL**

Dot Sentinel AI is an innovative cybersecurity dashboard designed to simplify the analysis of .PCAP log files using agentic AI, enabling rapid detection of network threats like malware, unauthorized access, or anomalies. It autonomously processes complex packet data, generating clear, user-friendly reports with intuitive visualizations that cater to both technical and non-technical users. The platform provides prioritized, actionable recommendations—such as blocking malicious IPs or updating protocols—to resolve identified issues efficiently.

With scalable, secure processing, Packet Sentinel AI empowers organizations to strengthen their cybersecurity posture, reduce response times, and protect critical assets effectively.



Objective



Purpose: To build an intelligent, end-to-end AI-powered Security Operations Centre (SOC) platform that automates network threat detection, analysis, and reporting.

Motivation:

- Traditional IDS systems cannot detect zero-day or evolving attacks
- Global shortage of 3.5 million cybersecurity professionals (ISC²)
- Alert fatigue and lack of actionable intelligence in existing tools
- Need for accessible, explainable security tools for SMEs and enterprises

Goal: Democratize real-time, intelligent, and explainable cybersecurity — making it accessible to users of all technical backgrounds.

Phase-1 Summary & Suggestions

What was presented in Phase 1:

- Initial system concept and architecture design
- Base ML models trained on UNSW-NB15 and NSL-KDD datasets
- Frontend prototype dashboard built in React.js
- Literature review and problem identification

Suggestions given by Supervisor/Reviewers:

- Integrate all modules into a single unified pipeline
- Improve frontend-backend API connectivity
- Add LLM-based reporting for human-readable output
- Include MITRE ATT&CK framework mapping
- Demonstrate end-to-end working with real PCAP input



**Next Review
Agenda**

Remedial Measures & Phase-2 Updates in the **Model**

Review-1 Suggestion

Unified pipeline

Frontend-backend connectivity

LLM-based reporting

MITRE ATT&CK mapping

End-to-end PCAP demo

Phase-2 Implementation

End-to-end PCAP → Detection → Report pipeline built

RESTful APIs + WebSocket live updates implemented

Gemma-3-1B-IT (Google) integrated for SOC

report generation

LLM agent auto-maps threats to ATT&CK techniques

Full pipeline deployed; PCAP Generator added for testing



**Model
Upgrades**

Existing Projects & Differentiation

Similar Existing Systems:

System	Approach	Limitation
Snort / Suricata	Signature-based IDS	Cannot detect zero-day threats
Splunk SIEM	Log aggregation & alerting	No AI reasoning or auto-reporting
Darktrace	ML anomaly detection	Expensive, closed-source, black-box
CrowdStrike Falcon	Endpoint detection	No PCAP-level network analysis

What makes Dot Sentinel AI different:

- Processes raw .PCAP files directly — no agent installation needed
- Combines supervised + unsupervised ML in a dual-layer engine
- Auto-generates human-readable SOC reports using LLM
- MITRE ATT&CK mapping built-in
- Open-source, deployable, and accessible to SMEs

PROJECT MODULES

Module 1 — PCAP Ingestion & Parsing

Accepts .PCAP files via drag-and-drop. Uses Tshark + Scapy to extract flow-level features: IPs, ports, protocols, timestamps, flags, packet sizes.

Module 2 — Dual-Layer Detection Engine

- Binary LSTM classifier → Malicious or Benign
- Random Forest multi-class classifier → Attack type (DoS, Exploits, Recon, etc.)
- Isolation Forest → Anomaly score for unknown/zero-day threats

Module 3 — Campaign Clustering

DBSCAN groups related alerts into unified attack campaigns, reducing alert fatigue.

Module 4 — Multi-Agent AI Layer

Gemini/Gemma LLM agents: map threats to MITRE ATT&CK, assign severity, generate SOC incident reports with mitigation steps.

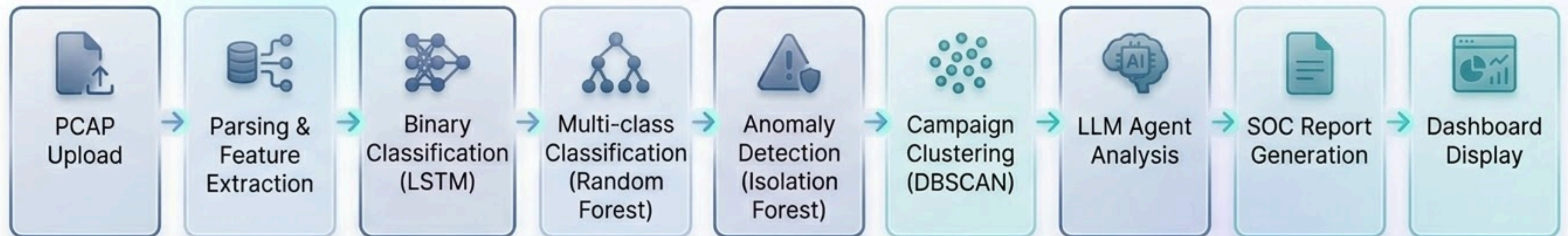
Module 5 — Dashboard (Frontend)

React.js + Tailwind CSS. Real-time alert table, protocol charts, MITRE heatmap, campaign view, AI report display.



SYSTEM ARCHITECTURE DIAGRAM

SOC-Based Intrusion Detection System Pipeline



OUTCOME

Technical Outcomes:

- Fully functional AI-powered SOC platform deployed on cloud (Vercel + Render)
- Dual-layer detection covering both known and zero-day threats
- Auto-generated, human-readable incident reports with MITRE ATT&CK context
- DBSCAN clustering reduces alert noise into coherent attack campaigns
- PCAP Generator enables testing without live network infrastructure

Impact on Public/Community:

- Makes advanced cybersecurity tools accessible to SMEs and non-expert users
- Reduces dependence on scarce, expensive cybersecurity professionals
- Speeds up incident response through automated analysis and reporting
- Bridges the gap between complex ML outputs and actionable security decisions
- Deployable as an open, scalable platform for academic and enterprise use

Analysis Complete

Total Flows: 131

Campaigns Detected: 6

Campaign 0 (Flows: 103)

AI SOC Analysis:

Okay, here's a SOC-style incident report based on the provided campaign data.

Incident Report

****Date:**** October 26, 2023

****Subject:**** Potential Lateral Movement Campaign – Campaign ID 0

****Severity:**** MEDIUM

1. Incident Description

This report details a potential lateral

Analyze Another PCAP

Analysis Complete

Total Flows: 131

Campaigns Detected: 6

1. Incident Description

This report details a potential lateral movement campaign detected within our network. The campaign, identified as Campaign ID 0, has been observed across 103 flows. The campaign's primary attack type appears to be low-probability, with an average attack probability of 0.544. The average anomaly score is -0.001, indicating a relatively low level of suspicious activity. The confidence distribution suggests a moderate level of certainty regarding the campaign's nature – 73% of analysts believe it represents a genuine threat, while 24% are unsure, and 6% are highly suspicious. The campaign's flow count is significant, suggesting a potentially broad and persistent attack.

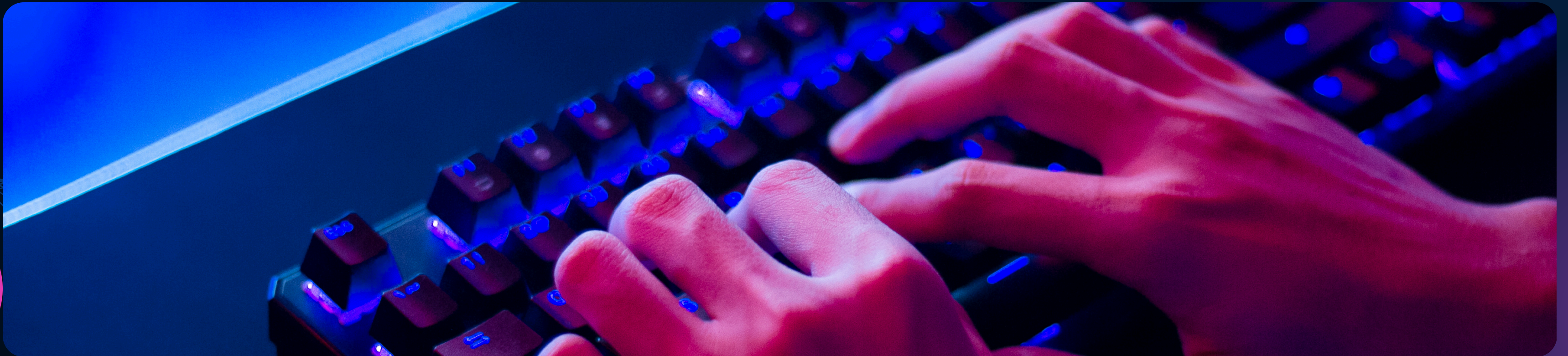
TEAM CONTRIBUTIONS

MEMBER	ROLE	KEY CONTRIBUTION
Siddhant Jha (22BCE10720)	Team Lead & Frontend Developer	React.js dashboard, WebSocket integration, MITRE heatmap, PCAP upload UI, system architecture design, team coordination
Prakhar Shukla (22BCE10247)	Backend Lead & System Architect	FastAPI backend, PCAP parsing pipeline (Tshark + Scapy), RESTful APIs, Docker containerization, modular architecture
Parthib Sarkar (22BCE10708)	Integration & Deployment Lead	Frontend-backend integration, cloud deployment (Vercel + Render), API configuration, environment setup, system testing
Nandini Sharma (22BCE10071)	ML Pipeline & Campaign Clustering	UNSW-NB15 preprocessing, LSTM + Random Forest models implementation, threshold tuning, DBSCAN clustering, event abstraction layer
Aashina Saxena (22BCE10114)	Anomaly Detection & LLM Integration	Isolation Forest implementation, anomaly scoring, Gemma LLM integration, SOC report generation, inter-module data flow

Conclusion

Group 08 thanks our supervisor, Dr. Sasmita Padhy for her guidance in making of this application so far. Her valuable suggestions helps us to make this project become success and without her, the project would be hard to complete. The application's upgrades after review 01 witnessed many drastic changes that were suggested and guided by her.

DOT SENTINEL had come closer to reality by another monumental step because of the collaboration of all 5 members of this group and because of each member's unique yet important skill set which was shown in today's meet.



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Thank You!

Group 08

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